

MUHAMMAD UMAIR ASGHAR

Metallurgy and Materials Engineer

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Current Address: Street 24, I 10/2, Islamabad

Permanent Address: Shorkot, Jhang



Objective

To seek a challenging and accelerated work experience in Metallurgy and Materials Engineering, leveraging technical expertise in Asset Integrity and Project Management to contribute to versatile projects and achieve organizational excellence.

Education

2021 - 2025	B.Sc. Metallurgy and Materials Engineering	Institute of Metallurgy and Materials Engineering, University of the Punjab, Lahore
2019- 2021	F.Sc. Pre-Engineering	Chenab College Shorkot

Work Experience

1. Bureau Veritas | Trainee Engineer Asset Integrity

02/2026 – Present | Islamabad

- Provide technical support to the AIM/RBI team in P&IDs and PFD Interpretation, data collection, data population, degradation mechanism identification, risk analysis, risk assessment, and inspection planning activities.
- Maintain accurate, organized and traceable documentation related to AIM/RBI processes and deliverables.
- Assist the team lead in ensuring AIM/RBI activities complied with applicable industry codes and standards, including API, NACE, ASME, and ASTM.
- Collaborate effectively with AIM/RBI team members to achieve project objectives, ensure timely project completion, and deliver high-quality services to clients.

2. Metron International | Project Manager – Effluent Flow Monitoring

09/2025 – 01/2026 | Lahore

- Liaise with clients, Board of management Sunder Industrial Estate, and regulatory bodies for data validation & reporting.
- Manage on-site operations involving installation and commissioning of electromagnetic flow meters, submersible pumps, and related instrumentation.
- Lead technical teams to conduct 24-hour flow monitoring, composite sampling, and data recording with strict adherence to quality and safety protocols.

3. Metron International | Proposal Engineer

07/2025 – 08/2025 | Lahore

- Liaised between sales and technical teams to generate comprehensive technical and commercial proposals for potential clients.

Internships

Velosi Integrity, Inspection, & Engineering Services

06/2024 – 07/2024 | Lahore & Peshawar

- Acquired practical knowledge in NDT Techniques (DPT, MPT, UT & RT), equipment calibration, and lifting inspection.

Chenab Engineering Works & Foundries Pvt Ltd

07/2023 – 08/2023 | Faisalabad

- Trainee Metallurgical Engineer across Drawing, Molding, Melting, Heat-treatment, Machining and Quality departments.

Personal Projects

- **Lignin derived carbon for Super capacitors electrodes.**
Developed sustainable carbon electrode materials from lignin (a paper industry by-product) for supercapacitor applications, focusing on material processing and electrochemical energy storage potential.
- **Laboratory-Scale Fabrication of Glass Fiber–Reinforced Composites (GFRC)**
Fabricated glass fiber–reinforced composite specimens at laboratory scale, emphasizing process optimization, fiber–matrix interfacial bonding, and mechanical strength enhancement.
- **Comparative Study of SMAW and GMAW Welding Techniques.**
Conducted a comparative study of Shielded Metal Arc Welding (SMAW) and Gas Metal Arc Welding (GMAW) using microstructural analysis, hardness testing, and Charpy impact testing to evaluate weld integrity and performance.
- **Corrosion Behavior of Aluminum Alloy 5083 in Saline Environment**
Evaluated the corrosion behavior of Aluminum Alloy 5083 under controlled saline conditions using weight-loss methods to determine corrosion rates and material degradation mechanisms.
- **Design and Casting of Hollow Cylindrical Components**
Designed and cast hollow cylindrical components with optimized riser and gating systems to improve casting quality, metal flow, and defect reduction.
- **Surface Characterization of Coated Mild Steel Samples**
Performed surface morphology and elemental composition analysis of coal dust and boron nitride (BN) coatings on mild steel using SEM and EDX techniques.

Volunteer Experiences

- Session moderator at 10th Invention to Innovation Summit 2025, University of the Punjab, Lahore
- Ambassador of Metacom 2.0, University of Engineering and Technology
- Presented Institute of Metallurgy & Materials Engineering at International Foundry Exhibition 2025, PC Hotel, Lahore.
- Participated as a debater in the Defense day competition at the Institute of Metallurgy and Materials Engineering (IMME), University of the Punjab, Lahore.
- Coordinator and Commentator in Annual Sports Gala held at (IMME), University of the Punjab, Lahore.

Achievements

- Secured 3rd position in Metallography competition at Metacom 3.0, University of Engineering & Technology (UET), Lahore.

Participations

- Metallography Competition at University of Engineering and Technology (UET), Lahore.
- Poster Competition at University of Engineering and Technology (UET), Lahore.
- Seminar on Business Intelligence & Optimization Solutions, University of the Punjab, Lahore.
- Ambassador of MetaCom 2.0 at University of Engineering and Technology (UET), Lahore.

Technical & Soft Skills

Microsoft Office, Communication and Interpersonal Coordination, Problem Solving, Resilient, Time management, Project management, Engineering drawing.

Interest & Hobbies

Travelling & Exploration, Book Reading, Badminton, Talking to elders.

Languages

English	Punjabi
Full Professional Proficiency	Native or Bilingual Proficiency
Urdu	
Native or Bilingual Proficiency	

References: References will be furnished on demand